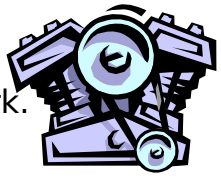
Energy Efficiency

When energy is changed some of it is always wasted. We can measure the amount of useful energy we get by the efficiency.

$$\text{Efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100$$

The examples below show energy changes in engines. The **work** that they do is their useful energy output.



Calculate the efficiency of each engine.

1. A petrol engine takes 1000J of chemical energy and does 250J of work.

2. A diesel engine uses 6000J of chemical energy and does 2100J of work.

3. An electric motor uses 50J of electrical energy and does 40J of work.

4. A cyclist (a human engine) has 200J of energy and does 30J of work.

For the following questions, you will need to decide what form the useful energy is in.

5. A TV takes in 600J of electrical energy and gives out 300J of light, 240J of sound and 60J of heat energy.

6. A windup toy uses 50J of potential energy and gives out 24J of kinetic energy, 20J of heat energy and 6J of sound energy.

7. 100J of electrical energy is put into a light bulb. 10J of light energy and 90J of heat energy are produced.

8. A hairdryer uses 750J of electrical energy and converts it into 400J of heat energy and 350J of sound energy.

9. 2000J of electrical energy is used by a microwave to produce 100J of sound energy, 400J of light energy and 1500J of heat energy.

10. An iPod uses 400J of energy which is transformed into 100J of light energy, 225J of sound energy and 75J of heat energy.